

VITORIA, Brazil

WMO: 836490

Lat: 20.258S

Lon: 40.286W

Elev: 3

StdP: 101.28

Time Zone: -3.00 (SAB)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.6	17.5	12.8	9.2	21.3	13.9	9.9	21.3	9.6	24.7	8.9	25.4	1.4	300	0.351

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	7.8	34.1	25.5	33.2	25.2	32.5	25.0	27.2	30.7	26.7	30.1	26.2	29.7	6.5	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.2	21.6	28.6	25.9	21.3	28.4	25.2	20.4	27.8	85.9	30.8	83.5	30.4	81.8	29.8	30.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.4	9.4	8.4	14.0	36.4	1.8	1.1	12.7	37.2	11.6	37.9	10.6	38.5	9.3	39.3		
(5)			13.3	28.2	1.4	1.1	12.3	29.0	11.5	29.6	10.7	30.2	9.8	31.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	25.5	27.9	28.3	27.7	26.4	24.4	23.3	22.8	23.1	24.0	25.2	25.6	27.0	(6)
(7)		DBStd	2.65	1.60	1.57	1.66	1.73	1.70	1.72	1.73	1.89	2.07	2.26	2.31	1.94	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD10.0	5643	556	512	548	492	446	400	397	406	420	471	469	528	(10)
(11)		CDD18.3	2601	297	278	289	242	187	150	138	148	170	213	219	269	(11)
(12)		CDH23.3	21114	3064	3011	2915	2139	1223	839	681	753	940	1464	1631	2452	(12)
(13)		CDH26.7	6514	1123	1181	1034	643	272	177	133	143	209	391	435	773	(13)
(14)	Wind	WSAvg	3.7	4.5	3.9	3.4	3.0	2.9	2.8	3.0	3.5	4.3	4.5	4.3	4.4	(14)
(15)	Precipitation	PrecAvg	1276	130	78	141	109	69	50	56	67	120	210	190		(15)
(16)		PrecMax	1974	299	239	426	395	247	175	127	211	163	445	649	687	(16)
(17)		PrecMin	667	10	10	14	18	1	0	4	6	21	26	40	18	(17)
(18)		PrecStd	334	81	53	100	77	56	40	33	43	44	88	137	126	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.8	35.2	34.9	34.0	32.8	31.8	31.1	31.1	32.2	33.2	34.0	34.3	(19)
(20)			MCWB	26.1	26.0	25.8	25.3	24.0	23.3	22.2	22.5	22.9	24.0	24.6	25.6	(20)
(21)		2%	DB	33.3	33.9	33.2	32.3	30.8	30.0	29.5	29.7	30.2	31.6	32.0	33.0	(21)
(22)			MCWB	25.4	25.6	25.4	25.1	24.0	22.8	22.0	22.1	22.5	23.9	24.5	25.3	(22)
(23)		5%	DB	32.4	32.9	32.3	31.1	29.2	28.8	28.1	28.2	29.0	30.2	30.8	31.9	(23)
(24)			MCWB	25.2	25.5	25.4	25.0	23.5	22.7	21.9	21.9	22.4	23.6	24.5	25.1	(24)
(25)		10%	DB	31.5	32.0	31.3	30.0	28.1	27.3	26.9	27.0	27.8	29.0	29.2	30.7	(25)
(26)			MCWB	24.9	25.2	25.4	24.7	23.2	22.4	21.7	21.6	22.0	23.2	24.1	25.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.0	28.0	27.6	26.9	26.0	25.3	24.2	24.4	24.9	25.7	26.3	27.1	(27)
(28)			MCDWB	31.9	32.6	30.8	30.1	28.4	28.3	27.5	27.9	29.1	29.5	30.5	31.1	(28)
(29)		2%	WB	26.8	27.1	26.9	26.3	25.3	24.2	23.4	23.6	24.0	25.0	25.7	26.4	(29)
(30)			MCDWB	30.4	30.7	30.3	29.6	28.2	27.3	26.6	27.0	27.1	28.7	29.7	30.2	(30)
(31)		5%	WB	26.2	26.5	26.4	25.8	24.7	23.6	22.8	22.9	23.4	24.4	25.1	25.9	(31)
(32)			MCDWB	29.8	29.9	29.8	29.2	27.7	26.7	25.9	26.1	26.6	28.0	28.7	29.6	(32)
(33)		10%	WB	25.8	26.1	26.0	25.4	24.1	23.0	22.3	22.4	22.9	23.9	24.6	25.5	(33)
(34)			MCDWB	29.3	29.4	29.4	28.6	26.9	26.2	25.5	25.6	26.1	27.4	27.9	29.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.3	7.8	7.5	7.3	7.6	8.2	8.3	8.0	7.3	7.0	6.3	6.8	(35)
(36)			MCDBR	8.6	8.9	9.0	9.0	9.7	10.5	10.7	9.8	9.1	8.8	8.4	8.6	(36)
(37)			MCWBR	3.2	3.2	3.3	3.6	4.5	4.9	4.8	4.3	3.7	3.4	3.3	3.4	(37)
(38)		5% WB	MCDWB	7.6	8.0	7.9	7.7	8.5	8.9	9.2	8.3	7.7	7.6	7.3	7.5	(38)
(39)			MCWBR	3.3	3.3	3.5	3.6	4.4	4.8	4.7	4.1	3.6	3.5	3.4	3.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.400	0.396	0.394	0.383	0.374	0.361	0.371	0.391	0.456	0.441	0.424	0.421	(40)	
(41)		taud	2.509	2.513	2.537	2.551	2.546	2.580	2.534	2.459	2.253	2.343	2.428	2.447	(41)	
(42)		Ebn at Noon	945	938	915	886	855	849	850	864	843	885	916	926	(42)	
(43)		Edh at Noon	115	113	106	97	91	85	91	105	138	132	124	122	(43)	
(44)	All-Sky Solar Radiation	RadAvg	6.37	6.34	5.56	4.73	3.92	3.66	3.82	4.46	4.80	5.27	5.21	5.75	(44)	
(45)		RadStd	0.87	0.54	0.46	0.30	0.22	0.26	0.26	0.28	0.50	0.61	0.64	0.72	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	-0.66	N/A	N/A	-0.66	-1.05	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page